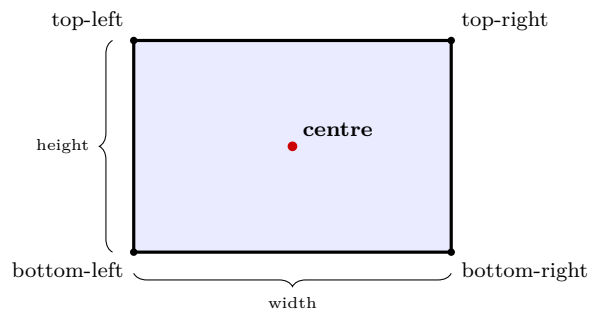


*Answer key*

**A point is a spot. A sprite is a box.** A point has two numbers: its  $x$  (how far across) and its  $y$  (how far up), written  $(x, y)$ . But a sprite in Scratch is not a single spot. It is a *box* that covers a patch of the grid. So how does Scratch know where the box sits? It pins the box to the grid by *one* point: the **centre**. When Scratch says a sprite is at  $(x, y)$ , it means the *centre* of the box is at  $(x, y)$ .

The box has other important points too. It has four **corners** and four **edges**. Each one has its own coordinates.



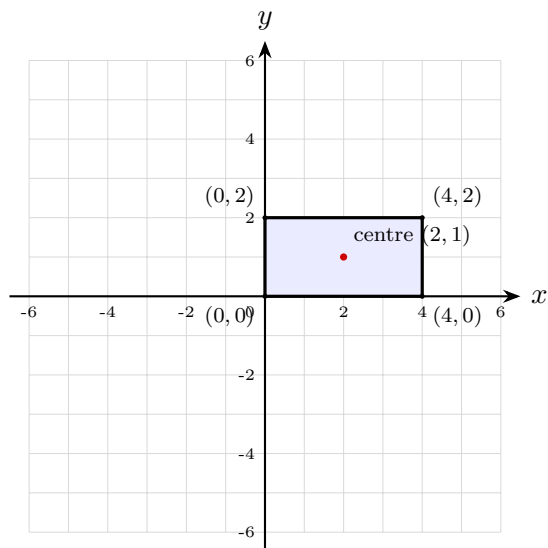
**Finding a corner from the centre.** A corner is half a width across and half a height up or down from the centre. So:

$$\text{corner } x = \text{centre } x \pm (\text{width} \div 2) \qquad \text{corner } y = \text{centre } y \pm (\text{height} \div 2)$$

**Why this matters for the Dino game.** To check if the Dino is standing on the ground, you must look at its *feet*: the bottom edge. You cannot look at the centre, because the centre sits higher up. The feet are at  $(\text{centre } y) - (\text{height} \div 2)$ . Scratch only tells you the centre with **y position**, so you subtract half the height yourself. Get this wrong and the Dino floats above the floor or sinks into it.

**Problem 1.**

**Read the box.** The box below has its centre marked. It is **4** wide and **2** tall. Write the coordinates of each part.

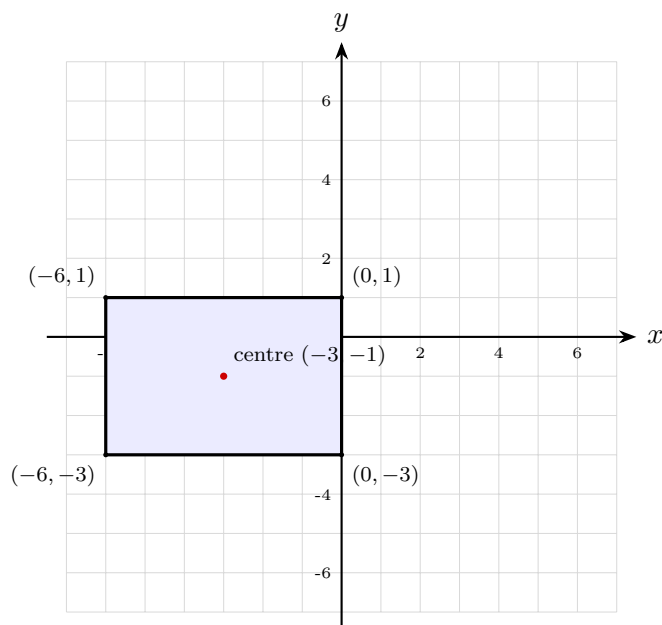


|                         |        |
|-------------------------|--------|
| the bottom-left corner  | (0, 0) |
| the bottom-right corner | (4, 0) |
| the top-left corner     | (0, 2) |
| the top-right corner    | (4, 2) |

## Problem 2.

**Read the box.** The box below has its centre marked. It is **6** wide and **4** tall. Write the coordinates of each part.

*Two of these are corners (two numbers each). Two are whole edges, so they need only one number.*



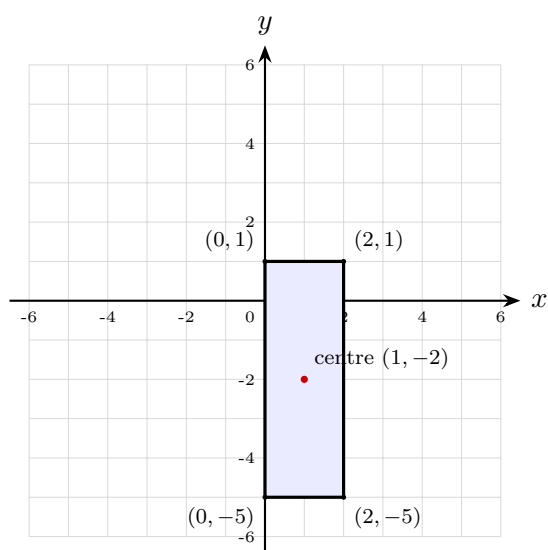
|                            |           |
|----------------------------|-----------|
| the top-left corner        | $(-6, 1)$ |
| the bottom-right corner    | $(0, -3)$ |
| the bottom edge (the feet) | $y = -3$  |
| the top edge               | $y = 1$   |

---

### Problem 3.

**Read the box.** The box below has its centre marked. It is **2** wide and **6** tall. Write the coordinates of each part.

*A tall, thin box. The left edge and the right edge each sit on one up-down line, so each has a single  $x$ .*

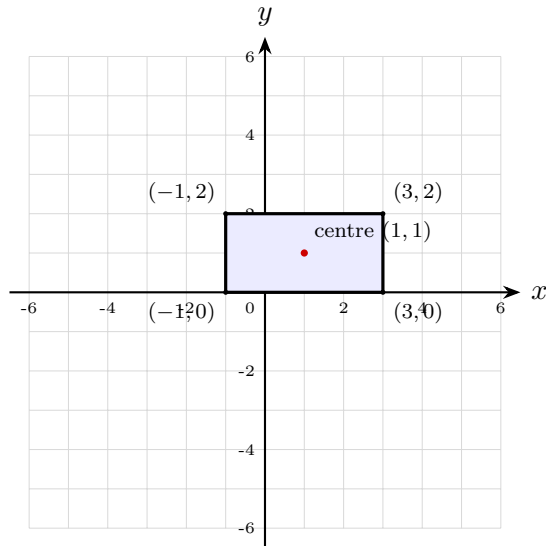


|                        |           |
|------------------------|-----------|
| the bottom-left corner | $(0, -5)$ |
| the top-right corner   | $(2, 1)$  |
| the left edge          | $x = 0$   |
| the right edge         | $x = 2$   |

---

### Problem 4.

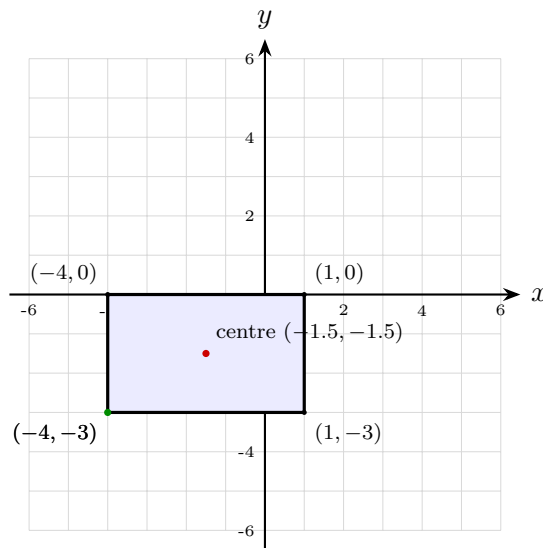
**Place the box.** Draw a box that is 4 wide and 2 tall, with its **centre** at  $(1, 1)$ . Mark the four corners.



### Problem 5.

**Place the box.** This time you are given a *corner*, not the centre. Draw a box 5 wide and 3 tall whose **bottom-left corner** is at  $(-4, -3)$ .

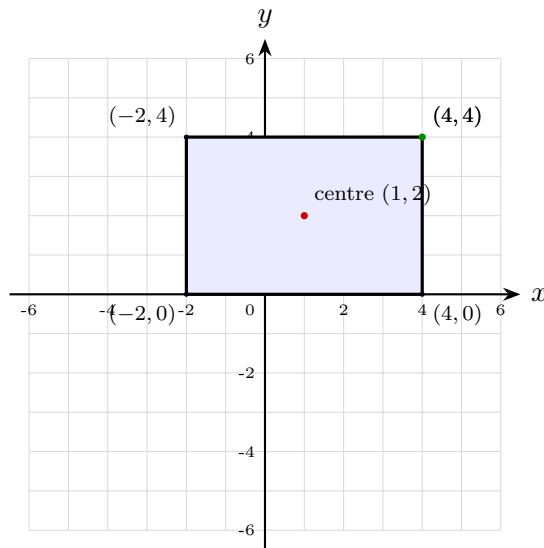
*Same kind of numbers as before, but the box lands in a different place. From the bottom-left corner the box goes to the right and up.*



*Notice the centre is not at  $(-4, -3)$  here. Choosing a different anchor point moves the whole box.*

### Problem 6.

**Place the box.** Draw a box 6 wide and 4 tall whose **top-right corner** is at  $(4, 4)$ . From the top-right corner the box goes to the left and down.



### Problem 7.

**Work it out.** A box has its centre at  $(0, 0)$ . It is **4** wide and **2** tall. There is no picture this time. Work out the coordinates using the rule above.

*Half the width is  $4 \div 2 = 2$ ; half the height is  $2 \div 2 = 1$ .*

|                            |            |
|----------------------------|------------|
| the top-right corner       | $(2, 1)$   |
| the bottom-left corner     | $(-2, -1)$ |
| the bottom edge (the feet) | $y = -1$   |

### Problem 8.

**Work it out.** A box has its centre at  $(5, 3)$ . It is **2** wide and **4** tall. There is no picture this time. Work out the coordinates using the rule above.

|                         |          |
|-------------------------|----------|
| the bottom-right corner | $(6, 1)$ |
| the top-left corner     | $(4, 5)$ |
| the top edge            | $y = 5$  |

### Problem 9.

**Work it out.** A box has its centre at  $(-2, 4)$ . It is **8** wide and **6** tall. There is no picture this time. Work out the coordinates using the rule above.

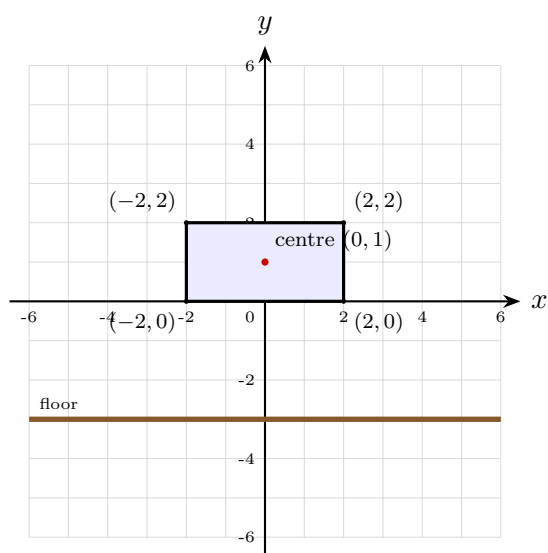
*A big box. Half the width is 4, half the height is 3.*

|                            |           |
|----------------------------|-----------|
| the bottom-left corner     | $(-6, 1)$ |
| the bottom-right corner    | $(2, 1)$  |
| the top-left corner        | $(-6, 7)$ |
| the top-right corner       | $(2, 7)$  |
| the bottom edge (the feet) | $y = 1$   |

---

### Problem 10.

**Above or below?** The brown line is the floor. Is the box above it, resting on it, or below it? Look at the feet (the bottom edge), not the centre.



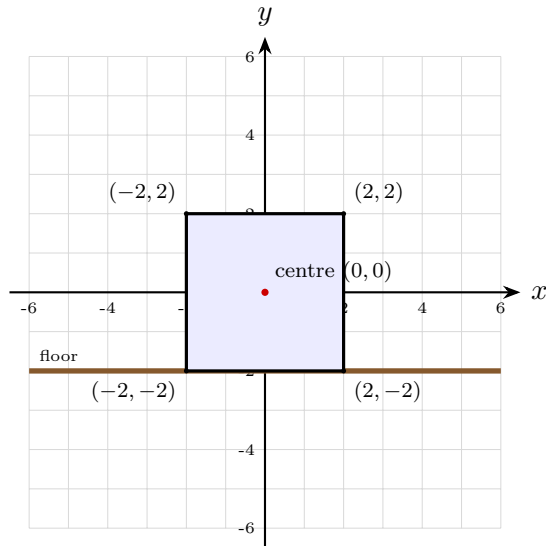
**Answer:** the box is **above the floor**.

The feet are at centre  $y$  minus height  $\div 2 = 1 - 1 = 0$ . The floor is at  $-3$ . Since 0 is greater than  $-3$ , the box is above the floor.

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### Problem 11.

**Above or below?** Is this box above the floor, on it, or below it? Work out where the feet are first.



**Answer:** the box is **resting on the floor**.

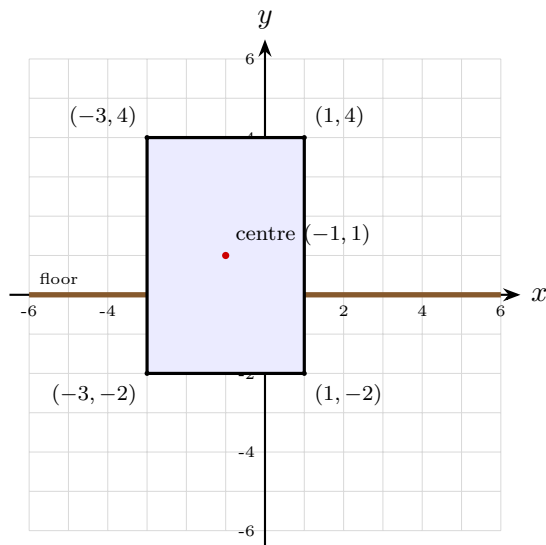
The feet are at centre  $y$  minus height  $\div 2 = 0 - 2 = -2$ . The floor is at  $-2$ . Since  $-2$  is equal to  $-2$ , the box is resting on the floor.

---

### Problem 12.

**Above or below?** And this tall box? Its centre looks like it is well above the floor. Check the feet before you decide.

*The centre can be above the floor while the feet are already below it. That is exactly the trap the Dino game sets.*



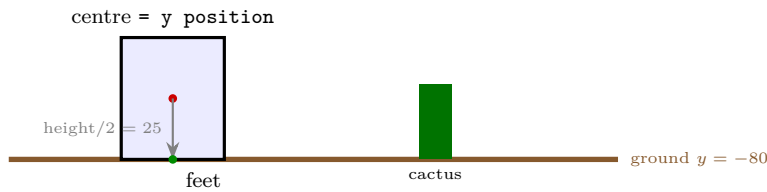
**Answer:** the box is **below the floor (sunk in)**.

The feet are at centre  $y$  minus height  $\div 2 = 1 - 3 = -2$ . The floor is at  $0$ . Since  $-2$  is less than  $0$ , the box is below the floor (sunk in).

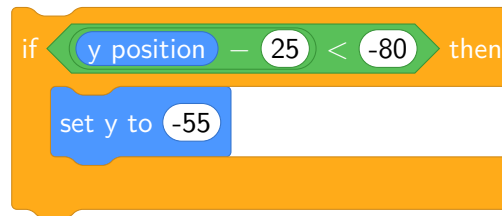
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### Problem 13.

**Write the block.** Now the real Scratch stage. The floor is at  $y = -80$ . The Dino is 50 tall, so  $y$  position minus 25 is its feet. Write the **if** that is true when the Dino has landed on the floor. When it lands, set its centre to  $-55$  so the feet rest on  $-80$ .



**The block:**



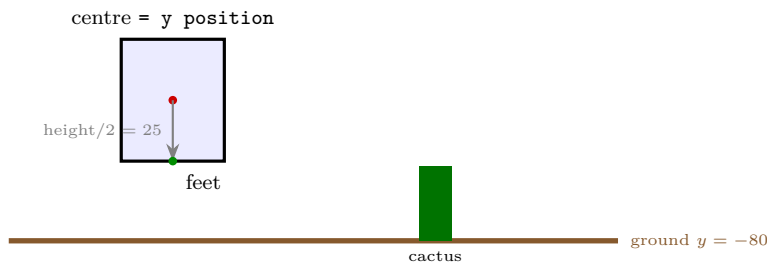
*The feet are  $y \text{ position} - 25$ . They reach the floor when that value drops to  $-80$ , so the test is  $(y \text{ position}) - 25 < -80$ . Setting the centre to  $-55$  puts the feet exactly on  $-80$  ( $-55 - 25 = -80$ ).*

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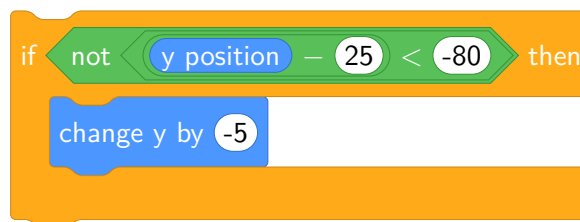
### Problem 14.

**Write the block.** While the Dino is jumping it is *above* the floor. Write the **if** that is true while the Dino is still in the air (its feet have *not* reached the floor yet). While it is, pull it down by 5.

*"Still in the air" is the opposite of "landed". You can wrap the landed test in **not**.*



**The block:**



*Landed is  $(y \text{ position}) - 25 < -80$ . Still in the air is the opposite, so wrap it in **not**. Pulling down by 5 each time is gravity.*

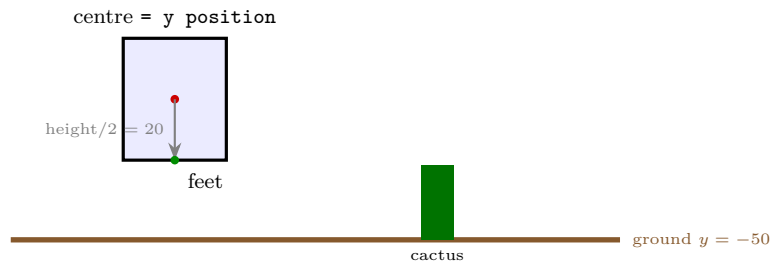
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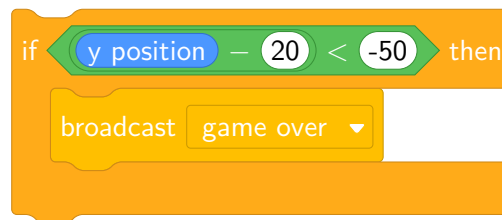
### Problem 15.

**Write the block.** A different check. The top of the cactus is at  $y = -50$ , and this Dino is 40 tall (so subtract 20 for its feet). Write the **if** that is true when the Dino's feet drop below the top of the cactus. If they do, broadcast **game over**.

*Same shape as before, but a different line ( $-50$ , not  $-80$ ) and a different height (subtract 20, not 25).*



**The block:**



*The feet are  $y \text{ position} - 20$ . Below the cactus top means less than  $-50$ :  $(y \text{ position}) - 20 < -50$ . You wrote the same pattern three times with different numbers. That is the whole grounded-check idea.*